User Manual

EEVblog 121GW multimeter with the open firmware F-1.06



This manual describes the open firmware for the EEVblog 121GW. It explains every key, every measuring function, every page of the SETUP menu and every message the meter can show. The pictures of the display were made with the firmware itself, so they show exactly what your meter shows.

The open firmware is not made or supported by EEVblog. Installing it is at your own risk (see the licence).

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Safety first

A multimeter is connected to things that can hurt you. Read this chapter before you use the meter.

> **WARNING** The meter is rated **600 V CAT III**. Never measure above that, and use only test leads and accessories rated at least as high as the meter.

- **Before you turn the dial** to another function, take the test leads off the circuit.
- **Before you measure**, check that the leads are in the right sockets for what you want to measure.
- **Check your leads** before every use. Replace damaged leads or probes.
- **Indoors and dry only.** The meter is not made for rain, moisture or dust.
- **Current measurements** go through fuses: the **A** socket takes up to 10 A (fuse 11 A / 1000 V, 10×38 mm), the **mA/µA** socket up to 400 mA (fuse 400 mA / 600 V, 6×32 mm). If a lead is in a current socket while the dial is on another function, the meter shows `LEAd` and beeps. Move the lead before you measure.
- **High voltage**: from about 30 V the lightning symbol lights up as a reminder.
- **Capacitors**: discharge a capacitor before you measure it.
- **Opening the meter** (batteries, fuses): take the leads off and switch the meter off first. Replace fuses only with the same type and rating.
- **Fuse test**: turn the dial to a voltage function and connect a lead between V Ω and the A (or mA) socket. A beep means the fuse is good.
- **Calibration mode** (MODE then REL while switching on) can ruin the meter's calibration and voids the factory calibration. Do not use it without calibration equipment.
- **AC voltage with a large DC part**: automatic ranging can pick the wrong range. Use manual range (RANGE key).

Getting to know your meter



The front of the meter: display, keys and dial.

The dial

The dial switches the meter on and chooses what you measure. Both ends of the dial are **OFF**. From left to right:

Dial position	What it measures	Press MODE to switch to
Low Z	Voltage with a low-impedance input (stops "ghost" voltages)	-
V	Voltage	DC → AC → AC+DC
mV / Temp	Small voltages, temperature	DC mV → AC mV → temperature
Hz ms %	Frequency, pulse width, duty cycle	Hz → ms → %
Ω	Resistance, continuity, diode, capacitance	Ω → continuity → diode → capacitance

mVA / VA	Power (voltage × current)	between the two power modes; the socket you use picks mVA or VA
µA / µVA	Small currents and power	DC µA → AC µA → two µVA power modes
A / mA	Currents	AC ↔ DC

Each dial position **remembers** the mode you chose with MODE, also after switching off.

The keys

Every key does one thing when you **press** it briefly and often another when you **hold** it for half a second (this manual writes `MODE long` for "hold MODE"). A short beep means "done", two short beeps mean "not possible now".

Key	Press	Hold
MODE	Next measuring mode of the dial position	Backlight on/off
RANGE	Manual range; each press: next range. In diode mode: 3 V / 15 V	Back to automatic range
HOLD	Hold the reading, then auto-hold, then off	-
REL	Relative reading on/off (zero the display)	AC: 1 kHz filter on/off. Current: store a zero offset
PEAK	AC: 1 ms peak capture on/off	Bluetooth on/off
MIN/MAX	Start recording minimum/maximum, then step MAX → MIN → AVG → all	Leave MIN/MAX
MEM	Look at the last stored readings	Start/stop logging to the SD card
SETUP	Next page of the SETUP menu (small display)	Change the page shown (it blinks); hold again to save

The display

The **large display** shows the reading. The **small display** above it shows a page of the SETUP menu: the temperature inside the meter, the battery, the date and time, or extra information for the mode you use (for example the frequency of an AC voltage). The **bar graph** under the reading follows quick changes. Symbols show what is active: AUTO (automatic range), HOLD, REL, PEAK, MIN MAX AVG, MEM, BT (Bluetooth), 1kHz, Apo (auto power-off), the battery and the lightning symbol for high voltage.

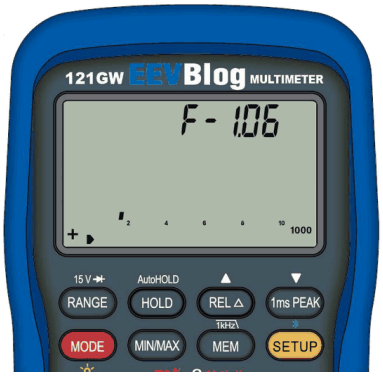
First steps

Batteries

The meter uses four AA batteries (1.5 V alkaline). The battery symbol shows when they get low (below about 4.1 V); replace them then. Lithium AA cells start at a higher voltage than alkaline cells. Remove the batteries if you will not use the meter for more than about 50 days. A small CR1220 battery inside keeps the clock running.

Switching on

Turn the dial from OFF to a function. The small display shows the firmware version for a moment, then the meter measures.



At power-on the small display shows the firmware version, `F-1.06`.

The meter switches itself off after 30 minutes without a key press (auto power-off). One minute before, it beeps five times; press any key to keep it on. You can turn auto power-off off in the SETUP menu (page `APOoF`).

Your first measurement

1. Black lead into **COM**, red lead into **V Ω**.
2. Turn the dial to **V**. The meter starts in DC volts (or the mode you used last).
3. Touch the probes to the circuit. The reading appears; **AUTO** means the meter picks the range itself.



Measuring DC volts.

Measuring

For every measurement: choose the function with the dial, the mode with **MODE**, then connect the leads.

Voltage (V, mV, Low Z)

V measures up to 600 V. **MODE** switches DC, AC and AC+DC. In AC the small display can show the frequency of the signal and its level in dBm (see the SETUP menu). **mV** is for small voltages up to 500 mV. **Low Z** loads the circuit a little, so voltages picked up from nearby wires ("ghost voltages") disappear; it chooses AC or DC by itself and shows dashes while it decides. Do not use Low Z below about 12 V.



AC volts; the small display shows the frequency.



Low Z.

Temperature

Dial to **mV / Temp**, press **MODE** twice and connect the thermocouple. The meter is set for a K-type thermocouple; for J, T, N, R or S types choose the type in SETUP (page `tc`). °C or °F is chosen on the first SETUP page.



Temperature with a thermocouple.

Frequency, pulse width and duty cycle

Dial to **Hz ms %**. `MODE` steps through frequency (Hz), pulse width (ms) and duty cycle (%).



Frequency.

Resistance, continuity, diode, capacitance

Dial to **Ω**. `MODE` steps through:

- **Resistance** up to 50 MΩ. Short the probes and press `REL` to remove the resistance of the leads.
- **Continuity**: the meter beeps below a threshold (normally 30 Ω). The threshold and a "beep when open" mode are on the small display: hold `SETUP` and choose with `REL` / `PEAK` (`DN 30` = beep below 30 Ω, `UP 30` = beep above 30 Ω, from 1 to 300 Ω).
- **Diode**: shows the forward voltage. `RANGE` switches between a 3 V and a 15 V test voltage (for LEDs and Zener diodes). The meter starts with 3 V again after a mode change.
- **Capacitance** up to 10 000 µF. Discharge the capacitor first; `REL` removes the capacitance of the leads.



Resistance.



Continuity.

Current (µA, mA, A)

Move the red lead to the right socket first: **µA/mA** for small currents, **A** for up to 10 A. On the **A / mA** position the meter sees which socket you use and shows mA or A by itself. `MODE` switches AC and DC.

> **NOTE** In DC mA the small display can show the reading as a 4–20 mA loop percentage (SETUP page `LP`).

Burden voltage: in the current and power modes the small display has a page `bd0FF`. Turn it on (`bd0n`) to see the voltage the meter itself drops. For that, connect a lead from the current socket in use to the V Ω socket.



DC current in mA.

Power (µVA, mVA, VA)

The power positions measure voltage and current at the same time and show their product (VA). Connect the circuit so that the COM socket is common to both the voltage and the current measurement. `MODE` switches between the two power modes of the position. After a change the display shows dashes until the first result.

Ranges

Normally the meter picks the range (**AUTO**). Press `RANGE` to keep the present range; each further press steps to the next range. Hold `RANGE` to go back to automatic. When the input is too large for the range, the display shows `OFL`.

Functions

HOLD and auto-hold

Press **HOLD** once: the reading freezes (HOLD). Press again: **auto-hold** (A-HOLD) waits until the reading is stable, keeps it with a beep and updates when you measure something new. It ignores readings near zero, so it waits while the probes are in the air. Press again: off. In Hz/ms/%, capacitance and during MIN/MAX the second press switches HOLD off (no auto-hold). HOLD does not work while logging.



HOLD keeps the reading.

REL: relative reading

Press **REL**: the present reading becomes the zero point and the meter shows the difference. Press again to switch off. With SETUP page `rEL-P` the difference is shown in **percent** instead, handy for checking parts against a reference.



REL shows the difference to the reference.

MIN / MAX and statistics

Press **MIN/MAX** to start recording. Each further press shows the maximum, the minimum, the average, and all three symbols. Hold **MIN/MAX** to stop. While MIN/MAX runs, the SETUP menu has four extra pages: `n` (number of readings), `Sd` (standard deviation, shown on the large display), `tHI` and `tLo` (the time of the maximum and the minimum).



The maximum while MIN/MAX runs.

PEAK

In AC volts and AC mV, **PEAK** captures fast peaks (1 ms). Press again to switch off.

1 kHz filter

In AC volts and AC mV, hold **REL** to switch a 1 kHz low-pass filter on or off (1kHz symbol). It reduces high-frequency noise on the reading.

Backlight

Hold **MODE** to switch the light on or off. The meter remembers your choice. On SETUP page `bL` you can let the light switch off by itself after 30 seconds or 1 to 99 minutes; any key or dial turn switches it on again (that first key press only wakes the light).

Zero offset for currents

In a current mode on its lowest range, hold **REL** with nothing connected to store the present small reading as a zero offset. It is kept also after switching off. Each time you do this the new offset is added to the old one, and there is no key to clear it, so use it only when the meter really shows a small current with nothing connected.

The SETUP menu

The SETUP menu lives on the small display. You can use it while you measure, but not during logging or while you look at stored readings.

How to use it

1. Press **SETUP** to go to the next page. The pages go round in a circle.
2. To change a page, hold **SETUP** until it blinks.
3. Change the value with **REL** (up / next) and **PEAK** (down / previous).
4. Hold **SETUP** again to **save** (long beep). A short **SETUP** instead leaves without saving.

Saved settings stay after switching off. If you leave a page blinking for about two minutes, the change is undone. Some pages only appear in certain modes (they are marked below).

Pages for every mode



Temperature inside the meter (°C / °F)

Shows the meter's own temperature. Hold SETUP and press REL to switch between °C (`c`) and °F (`F`). This also sets the unit of the temperature mode.



Battery `BAT`

Shows the battery voltage. To correct a meter that reads the battery a little high or low, hold SETUP: `bA` and the voltage blink; REL / PEAK change it in 0.2 % steps (up to ±10 %); hold SETUP to save.



Auto power-off `APOoN` / `APOoF`

On: the meter switches off after 30 minutes without a key press. Off: it stays on until you turn the dial. It never switches off during logging or MIN/MAX.



Buzzer `b-On` / `b-OFF`

Off silences key beeps and the other beeps. The continuity beep and the warnings (wrong socket, high voltage) always sound.



Contrast `Lcd-4`

Display contrast from 0 to 7. Turn it down if you see faint "ghost" segments, up if the display is too pale.



Date and time

Three pages: the year (`2026`), month and day, hour and minute. Month and day are set in two steps: hold SETUP to save the month, then the day blinks; hold SETUP again. Hour and minute work the same way. The clock is used for log files. The companion app can also set it from the internet.



Serial number (5 digits)

The meter's ID, written into log files. Hold SETUP; the left digit blinks. REL / PEAK change the blinking digit, a short SETUP moves to the next digit. Hold SETUP to save.

> **WARNING** A calibration backup (cal.bin) can only be restored to the serial number it was made with. If you change the serial number, make a new backup afterwards.



Logging interval `In`

How often a reading is logged, 0 to 999 seconds. 0 = about five readings per second.



Backlight timer `bL`

`bLOFF` = no timer (the light stays as you switch it with MODE hold), `bL30S` = 30 seconds, `bL 1` to `bL 99` = minutes.



Bluetooth speed `bt`

How many readings per second go out over Bluetooth: 2 (as the original), 4, 10 or 20. Use 10 or 20 for live graphs, 2 to save battery.



Bluetooth transfer `Ld-bt`

The page for the companion app: firmware updates, the SD card files and the settings over Bluetooth. Readings pause while this page is shown. See the Bluetooth chapter.



Format the card `Fr no`

Only with the two-part bootloader. Erases the SD card and formats it as exFAT. See the SD card chapter.



REL display `rEL-d` / `rEL-P`

d = REL shows the difference, P = REL shows the difference in percent of the reference.



Display level `dSP 1` / `dSP 2`

1 = calm, steady digits (as the original). 2 = the digits follow changes about four times faster, with a little more flicker.



Clock display `CL 24` / `CL 12`

24-hour or 12-hour time on the meter. Log files always use 24-hour time.



Limit test `Lt`, `Lo`, `HI`

Sorts parts into PASS, too high or too low:

1. Put a part with the lowest allowed value on the probes, go to `Lo`, hold SETUP, press REL (the reading is taken), hold SETUP.
2. Do the same with the highest allowed value on page `HI`.
3. On page `Lt` hold SETUP, REL to `LtOn`, hold SETUP.

The small display now shows `PASS`, `HI` or `Lo` for each part, with a beep when it leaves PASS. Both limits must be taken in the same mode and range; the test cannot be switched on before. While it runs the range stays fixed. Turning the dial or changing the mode ends the test. The limits are forgotten at power-off.



Logging trigger `tr`

`tr-In` = log every interval. `tr-Ch` = log when the reading changes by at least the amount on the next page `Ch` (1 to 255 steps of the last digit). `tr-Lt` = log when the limit test result changes. With an interval of 0 the meter logs only on the trigger; with a larger interval it also logs at every interval.

Pages for certain modes

Page	Mode	What it does
Frequency, then dBm	AC volts	Two pages: the frequency of the signal, then its level in dBm. Hold SETUP on the dBm page to choose the reference (`r 600`, 50 Ω, 75 Ω or dBV).
`tc-K`	Temperature	Thermocouple type K, J, T, N, R or S.
`DN 30`	Continuity	Beep threshold 1–300 Ω; DN = beep below, UP = beep above.
`bH100`	Continuity	How long the beep holds after a short contact: 100, 50 or 25 ms, or `bHOFF` = shortest (about 15 ms).
`3` / `15`	Diode	The test voltage (change it with RANGE).

`LPOFF`, `L4–20`, `L0–20`	DC mA	Show the current as a loop percentage: 4 mA = 0 %, 20 mA = 100 % (or 0–20 mA).
`PtOFF`, `Pt100`, `P1000`	Resistance	Show a PT100 or PT1000 sensor as a temperature. For best accuracy short the leads, press REL, then connect the sensor.
`bdOFF` / `bdOn`	Current and power	Show the burden voltage.
`n`, `Sd`, `tHI`, `tLo`	While MIN/MAX runs	Number of readings, standard deviation, time of the maximum and the minimum.

Logging and the SD card

Logging readings

1. Put a micro SD card into the slot under the battery cover.
2. Choose the interval (SETUP page `In`) and, if you like, a trigger (page `tr`).
3. Hold **MEM**: logging starts, MEM blinks and the small display counts the readings.
4. Hold **MEM** again to stop.



Logging: the small display counts the readings.

While logging, most keys are locked so nothing changes by accident; only backlight and Bluetooth still work. Auto power-off waits until logging stops. With low batteries logging does not start (`LoBAt`).

Each session is a file named after the date, for example **26110500.CSV** (2026-11-05, file 00). It is a plain text table that opens in any spreadsheet: date, time, function, value and unit per line, and the maximum and minimum with their times at the end.

> **TIP** Readings are written to the card in blocks. Stop logging before you switch off or take the card out, so the last readings are saved.

Looking at stored readings

Press **MEM** to see the last readings (up to 60). **REL** = next, **PEAK** = previous, **MEM** = back. The view closes by itself after about 10 seconds. If nothing has been logged yet, MEM gives two short beeps.

Cards

Any normal micro SD card works. Cards up to 32 GB usually come formatted as FAT32, larger ones as exFAT; the meter uses both.

> **NOTE** EEVblog's original bootloader can only read FAT32 cards for firmware updates. With the two-part bootloader exFAT works too.

Formatting a card on the meter

Only with the two-part bootloader. **Everything on the card is erased.**

1. SETUP to page `Fr no` (after `Ld-bt`).
2. Hold SETUP (`Fr no` blinks), press REL for `Fr yE`, hold SETUP.
3. `Fr --` while it works, then `Fr oK`.

Bluetooth and the Companion app

Hold **PEAK** to switch Bluetooth on (BT symbol); hold it again to switch off. The meter keeps measuring and sends its readings. Only one device can connect at a time: close the phone app before you use the computer app.

The **121GW Companion** is an app for Windows, macOS and Linux (see the project's releases; free for individuals, schools and non-profits, companies need written permission). It shows the meter live with a graph, records to your computer, sets the meter's clock, updates the firmware and the bootloader, and can show readings as percent, PT100 temperature or pass/fail. With the meter on SETUP page `Ld-bt` it can also:

- list, download and delete the files on the SD card, and copy new files into a folder on your computer;
- read and change most settings of the SETUP menu (not the logging interval, the serial number or the calibration).

Readings pause while `Ld-bt` is shown. A download runs at about 1 KB per second.

Backup of the calibration

Your meter's calibration, serial number and settings are stored inside the meter. Make a backup on the SD card, so you can restore them if something ever goes wrong.

Make a backup (cal.bin)

1. Card in the meter, dial to OFF.
2. Hold **MEM** and switch the meter on. Release MEM. The display shows `SdCAL`.
3. Within 5 seconds press **REL**: `SdSAv`.
4. Within 5 seconds press **MODE**. After a moment: `SAvEd`. Switch off.

If you wait too long, the meter simply starts normally; switch off and try again. `-Err-` means the backup did not work (no card, card problem).



`SdCAL`



`SAvEd`

The card now holds **cal.bin**. Copy it to your computer and keep it safe.

Restore the backup

Same as above, but press **PEAK** instead of REL (`SdSEt`), then **MODE**, each within 5 seconds. The meter only accepts the backup of **its own** serial number; another meter's file shows `-Err-` and nothing is changed.

> **WARNING** Restoring replaces the calibration, the serial number and all settings with those in the file. Use only your own meter's backup.

Serial number and date code

Log files start with an ID made of a four-digit date code and the five-digit serial number. The serial number is a page of the SETUP menu. The date code can only be stored when you hold **SETUP** while switching on: the year and month pages then also store it. Those pages also set the clock, so set the clock again afterwards. Normally you never need this.

Keys while switching on

Hold or press while switching on	What happens
MEM	Backup / restore menu (`SdCAL`), see above
MEM + HOLD	Firmware update from the card (bootloader)
SETUP	The year and month pages also store the date code
RANGE	All display segments on (display test)
MODE	`CAL1` appears. Do **not** press REL then; switch off. REL would start calibration mode.

Updating the firmware

Your calibration, serial number and settings are kept through every firmware update. Make a backup (cal.bin) first anyway, and use fresh batteries.

Which bootloader do you have? If your SETUP menu has the page `Fr no` after `Ld-bt`, your meter has the two-part bootloader. Otherwise it has EEVblog's original one.

With the Companion app (two-part bootloader)

When an update is available, the app offers it. Click **Update**: the meter restarts, shows `-IAP-` and the progress, then `Ld oK`, and restarts into the new firmware. Nothing to press on the meter.

With an SD card

1. Use fresh batteries.
2. Copy the firmware file to the card's top folder, named **EEVBlog.bin**. (EEVblog's bootloader needs a FAT32 card.)
3. Dial to OFF, hold **MEM** and **HOLD** and switch on.
4. The meter shows IAP (`-IAP-` with the two-part bootloader). Press **SETUP**: within about 8 seconds with EEVblog's bootloader, within 30 seconds with the two-part bootloader.
5. Wait until the meter restarts by itself. Do not take the card or the batteries out until then. The two-part bootloader shows `douun` with the bar graph, then `Good`; `FAIL` means the file was refused and nothing was changed.

You can also send the firmware file to the card over Bluetooth: Bluetooth on (hold PEAK), SETUP page `Ld-bt`, then send it with the companion app. The meter shows `Ld` and the percent, then `Ld oK`: the file is now on the card. With EEVblog's bootloader you then do the update from the card (steps 3 to 5 above); with the two-part bootloader the app does that for you.

The two-part bootloader

The two-part bootloader is an optional replacement for EEVblog's bootloader (the small program that installs firmware). With it, updates go over Bluetooth without a card or keys, cards may be exFAT, and an interrupted update is simply repeated. It never touches the calibration. The companion app installs it once; you need fresh batteries for that. Details: the bootloader guide on the project page.

Your calibration, serial number and settings are kept through every firmware update.

Messages

Display	Meaning	What to do
`OFL`	Input too large for the range (or outside the sensor range)	Choose a higher range or reduce the input
`LEAd` + beeping	A lead is in a current socket while the dial is on another function	Move the lead to V Ω, or turn the dial to the current function
`-----`	No value yet (power after a change, percent without reference, statistics with too few readings)	Wait or measure
`-oFF-`	Auto power-off is switching the meter off	To use the meter again, press a key (release it and press again)
`LoBAt`	Batteries too low for logging	Replace the batteries
`SdEnd`	The log file is full	Start a new log
`SdE01` …	SD card problem while logging (no card, cannot write)	Check the card, its free space and that it sits correctly
`SdCAL`, `SdSAv`, `SdSEt`	Backup menu at power-on	See "Backup of the calibration"
`SAvEd`	Backup or restore done	Switch off
`-Err-`	Backup or restore failed (no card, wrong file, another meter's file)	Check the card and the file
`Ld nn`, `Ld oK`	Bluetooth transfer running / finished	-
`LdE 1` … `LdE 6`	Transfer failed (card error or damaged file)	Send again
`LdE 7`	The bootloader cannot be installed on this meter	Nothing was changed
`LdE 8`	Batteries too low to install the bootloader	Fresh batteries
`LdE 9`	Installing the bootloader failed	Send it again
`Fr oK`, `FrE n`, `FrE L`	Format done / failed / not possible while logging	Try another card / stop logging
`-IAP-`	Update mode of the two-part bootloader	Send the firmware with the app, or use a card
`douun`, `Good`, `FAIL`	Card update by the two-part bootloader: running / done / file refused (nothing changed)	For FAIL: check the file EEVBlog.bin
`CAL1`	MODE was pressed while switching on	Switch off. Do not press REL
`CAL`	Calibration mode (you pressed MODE then REL at power-on)	Switch off and on. Do not use it unless you have calibration equipment

Troubleshooting

Problem	Solution
The meter does not react to a key	Two short beeps mean "not possible now", for example during logging. Stop logging first.
`SdCAL` does not appear, or MODE does nothing after `SdSAv`	Hold MEM while switching on, then press the next keys within 5 seconds each.
Bluetooth does not connect	Hold PEAK until BT shows. Close the phone app; only one device can connect.
The reading jumps between ranges	Press RANGE to keep one range.
The display looks faint or shows ghost segments	Change the contrast on SETUP page `Lcd`.
The light goes off by itself	SETUP page `bL`: set `bLOFF`.
The meter switches off	Auto power-off: SETUP page `APOoF`.
A card update with EEVblog's bootloader does nothing	The card must be FAT32 and the file named EEVBlog.bin in the top folder.
After an interrupted update the meter shows `-IAP-`	Send the firmware again with the app or use a card; the meter stays in update mode until it has a good firmware.

Measuring ranges

Function	Ranges
DC and AC volts	5 V, 50 V, 500 V, 600 V
DC and AC millivolts	50 mV, 500 mV
Low Z	600 V
Resistance	50 Ω, 500 Ω, 5 kΩ, 50 kΩ, 500 kΩ, 5 MΩ, 50 MΩ
Continuity	500 Ω, beep threshold 1–300 Ω
Diode	3 V or 15 V test voltage
Capacitance	10 nF, 100 nF, 1 µF, 10 µF, 100 µF, 10 000 µF
Frequency	99.999 Hz to 999.99 kHz
Pulse width	10 ms, 100 ms, 1000 ms
Duty cycle	to 99.0 %
DC and AC µA	50 µA, 500 µA
DC and AC mA	5 mA, 50 mA
DC and AC A	500 mA, 5 A, 10 A
Temperature (K)	–200 to 1350 °C; other thermocouple types have their own limits
Power	250 µVA to 500 VA

For the accuracy of each range see the specifications in EEVblog's original 121GW manual; the firmware does not change the meter's hardware.